

Genetic Change: Module 6

Mutation:

- Genes
 - Influence
 - Appearance
 - Behaviour
 - Susceptibility to illness
- No 2 people
 - Exactly the same
 - Due to genetic 'variations'
- Variations:
 - Caused by
 - Combined effect of
 - Genetics + environment

- Mutations

- Changes in dna

- Put in
 - New alleles
 - Into population

- Molecular level

- Change in
 - dna sequence

- Result from

- Spontaneous replication errors
- Environmental factors
 - Uv light
 - Cigarette smoke

- May change gene information

- Affects
 - Resulting protein
 - Or rna
 - Made using gene

- Vary in size
 - Single base changes
 - Large structural changes

- Affecting
 - Multiple genes
 - Multiple chromosomes
- Mutated genes
 - Different alleles of a gene
- Some genes
 - Have markers
 - Linked to mutations
 - Useful for identifying
 - Health risks

Mutagens:

- Are environmental agents
 - That
 - Alter dna
 - Cause mutations
- Process
 - Is called mutagenesis
 - (altering dna + causing mutations)
- Some mutagens
 - Are carcinogenic
 - (cancer-causing)
- Especially carcinogenic when
 - They affect genes
 - That control cell cycle
 - E.g.
 - Proto Oncogenes
 - Tumour suppressor genes

Chemical Mutagens:

- Cause mutations
 - When cells are
 - Exposed frequently
 - Or for prolonged periods
- Can
 - Change dna
 - Disrupt
 - Protein function

- Cellular processes

- Examples:
 - Ingested chemicals:
 - Alcohol
 - Tobacco bar
 - Some medications
 - charred/fatty foods
 - Food additives
 - Nitrites
 - Environmental irritants/poisons:
 - Benzene
 - Solvents
 - Pesticides
 - Asbestos

How they get incorporated into DNA:

- Chemical mutagens

- Have similar structure
 - To normal nitrogenous bases
 - In dna
 - ATGC
- Due to similar structure
 - Dna polymerase
 - May by mistake
 - Put in these mutagens
 - into dna strand during replication
- Faulty chemicals
 - 'Base analogs'
 - Look like real bases
 - Don't pair correctly

Common Types:

- Alkylating Agents:
 - Chemicals add alkyl groups
 - To dna bases
 - E.g.
 - Methyl or ethyl
- Deaminating agents:

- Remove an amino group (-NH₂)
 - From dna bases
 - Converting them into different bases
- Intercalating agents
 - Flat, planar molecules
 - That insert themselves
 - Between base pairs
 - In dna double helix

Environmental Irritants or Poisons:

- Chemically modify dna

- Damage

- Alter

dna

- Indirectly cause dna mutations

- Benzene:
 - Metabolised in body
 - Into reactive intermediates
 - Intermediates cause
 - Dna strand breaks
 - Chromosomal aberrations
 - Chromosomal deviations from normal type
 - Linked to
 - Leukaemia
 - Other blood cancers
- Organic solvents:
 - E.g.
 - Toluene
 - Formaldehyde
 - High reactive chemicals that
 - modify dna bases
 - Interfere with
 - Dna repair enzymes

- Can lead to

- Base modifications

- Faulty replication

- Pesticides and coal tars:
 - Have

- Hydrocarbons
- Heavy metals
- Substances can
 - Make dna adducts
 - Bind to
 - Or alter structure
 - Of dna chemically
- Increase risk of mutation
 - + cancer
 - With long term exposure

Asbestos:

- Not chemically similar
 - to dna bases
- Cause
 - Chronic inflammation
 - When fibres embed in tissues
 - Lungs
- Inflammation
 - Releases ROS
 - (reactive oxygen species)
 - Chemically damage dna
 - harm
 - Dna repair processes
- Associated with
 - Mesothelioma
 - Other lung cancers

Naturally Occurring Mutagens:

- In environment under normal conditions
- Can occur from
 - Biological sources
 - Virus
 - Bacteria
 - Fungi
 - Products of metabolism
 - In cells
 - Non biological sources

- Metals

- Mercury
- Cadmium

Biological Mutagens:

- End products of
 - metabolism
- By products made during
 - Metabolic processes in
 - Plants
 - Animals
 - Fungi
- E.g.:
 - Nitrosamines
 - Made in stomach when
 - Nitrous acid/nitrites
 - Add with amines
 - Especially made when
 - Food cooked at high temps
 - Frying
 - Grilling meats

- Transposons

- 'Jumping genes'

- Sections of dna
 - That spontaneously fragment
 - + relocate
 - + multiply
 - In genome
- When in chromosomal dna
 - Disrupt dna functioning
- May
 - cause frameshift mutations
 - Disrupt coding seqs.
 - activate/deactivate genes
 - Trigger cancer if
 - on
 - Proto Oncogenes
 - tumour suppressor genes

- Microbes:
 - Naturally occurring
 - Biological
 - Include
 - Virus
 - Bacteria
 - May directly alter dna in cells
- Virus:
 - Hepatitis b virus
 - rubella virus
 - Insert own genes in host dna
 - Interfere with normal gene expression
 - Lead to
 - Uncontrolled cell division
 - Inactivate regulatory genes
- Bacteria:
 - Helicopter pylori
 - Infect stomach lining
 - Cause chronic inflammation
 - Inflammation
 - Release free radicals (ros)
 - Damage dna
 - Weaken dna repair mechs
 - → mutations
 - Linked with
 - Stomach ulcers
 - Gastric cancer
- Effects:
 - Mutagenic microbes:
 - Put in their own base seq.
 - Into dna
 - ∴ change functioning of genes
 - + trigger cancers
 - Some bacteria
 - Cause inflammation where
 - Free radicals
 - (ROS) made

- Cause dna damage
- Reduce efficiency of
 - dna repair systems
- ∴ increase mutation
- Nitrosamines
 - Carcinogenic
- Can be put into dna
- + React chemically
 - To cause mutations
 - in stomach, intestinal cells
- Common in
 - Processed meats
 - Smoked meats
 - Sausages

Physical Mutagens:

- Heat
- Ionising radiation
 - Enough energy
 - To break chemical bonds in
 - Molecules
 - + dna
- Direct heat
 - Often combined with
 - Chemical
 - + naturally occurring mutagens
- Radiation
 - Transfer of energy
 - From source
 - Ionising radiation is harmful

Ionising Radiation:

- Short wavelength
- High energy radiation
- Uv
- X rays
- Gamma rays
- Enough energy to

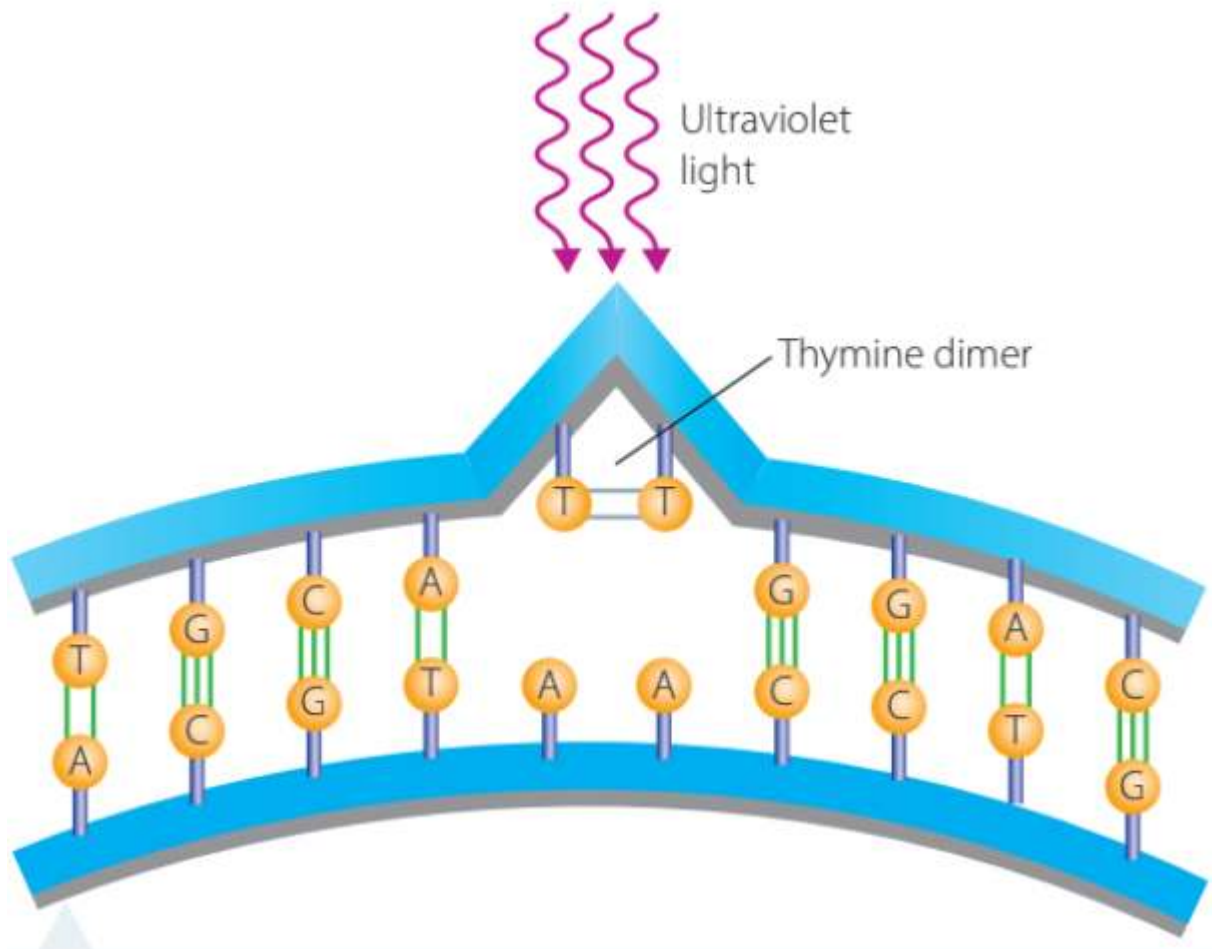
- Ionise atoms or molecules
- (Remove electrons)
- Damage dna
- Effects:
 - Leads to
 - Mutations
 - Chromosomal damage
 - Cancer
 - Direct:
 - Gets rid of electrons
 - From dna atoms
 - → dna ionisation → strand breakage
 - Indirect
 - Radiation interacts with
 - Water in cells
 - → forming free radicals
 - Free radicals
 - Highly reactive
 - Damage
 - Dna
 - Protein
 - Lipid
 - Other cell parts
- Radio waves + infrared
 - Low energy
 - Not harmful
- Uv
 - Moderate energy
 - Somewhat harmful
- Gamma rays
 - Very high energy
 - Extremely harmful
 - Even in small doses
- Other forms:
 - Cosmic rays
 - Radioactive materials
 - Nuclear testing/power plants

Electromagnetic Radiation:

- From sun
- Includes:
 - Radio waves
 - Microwaves
 - Infrared
 - Visible light
 - Uv
 - X rays
 - Gamma rays

Ultraviolet (UV) Radiation:

- Naturally occurring
- Uv radiation from sun
 - → skin cancer
- Uv light
 - Causes
 - making of pyrimidine dimers
 - (cross linked nucleotides)
- Pyrimidine dimers Happen when
 - Adjacent
 - Thymine or cytosine bases
 - On 1 strand
 - Get covalently bonded
- Prevents
 - Correct base pairing with
 - complementary strand
- Disrupts
 - Dna
 - Replication
 - Transcription
- ∴ affects
 - Gene expression
 - Cell cycle



Types of Mutations:

- 5 ways to categorise mutations:
 1. Cause of mutation
 - Spontaneous or induced
 2. Amount of gene change
 - Point mutation
 - Frameshift mutation
 - Also point mutation
 - Chromosomal mutation
 3. Effect of mutation on dna
 - Substitution
 - Insertion
 - Or deletion
 - Of nucleotide base
 - Can cause change on
 - 1 or many amino acids
 - No change in amino acids
 4. Effect of mutation on phenotype

- No change (silent)
 - Harmful
 - Beneficial
 - neutral
5. Heritability of mutations
- Does mutation happen
 - In somatic cell
 - Or germline (reproductive) cell

Point Mutation:

- 1 nucleotide variation
- May have
 - Significant effect
 - phenotype
 - If on gene exon
 - Gene expression
 - If on gene intron
- Named
 - According to
 - How they change
 - nucleotide seq
 - Of dna

Effect of Point Mutations on DNA:

- Change in base seq
 - Affects expression of
 - Mutated genes
- Most are
 - Base substitutions
- Base substitutions:
 - 1 nucleotide
 - Replaced by another
 - E.g.
 - C → T, G, A
- Changes lead to
 - Different amino acid added
 - During protein making
 - changes

- protein function

Example - Sickle Cell Anaemia:

- Cause
 - Point mutation in
 - Haemoglobin gene
 - CTC → CAC
 - Or GAG → GTG
- Result
 - Amino acid
 - Glutamate → valine
- Mutation changes
 - Haemoglobin shape
- Make red blood cells
 - Sickle shaped
- Abnormal cells
 - Block blood flow
 - In small vessels
 - Lead to
 - Pain
 - Fatigue
 - Organ damage
- Is Missense mutation
 - Amino acid change
 - Changes protein function

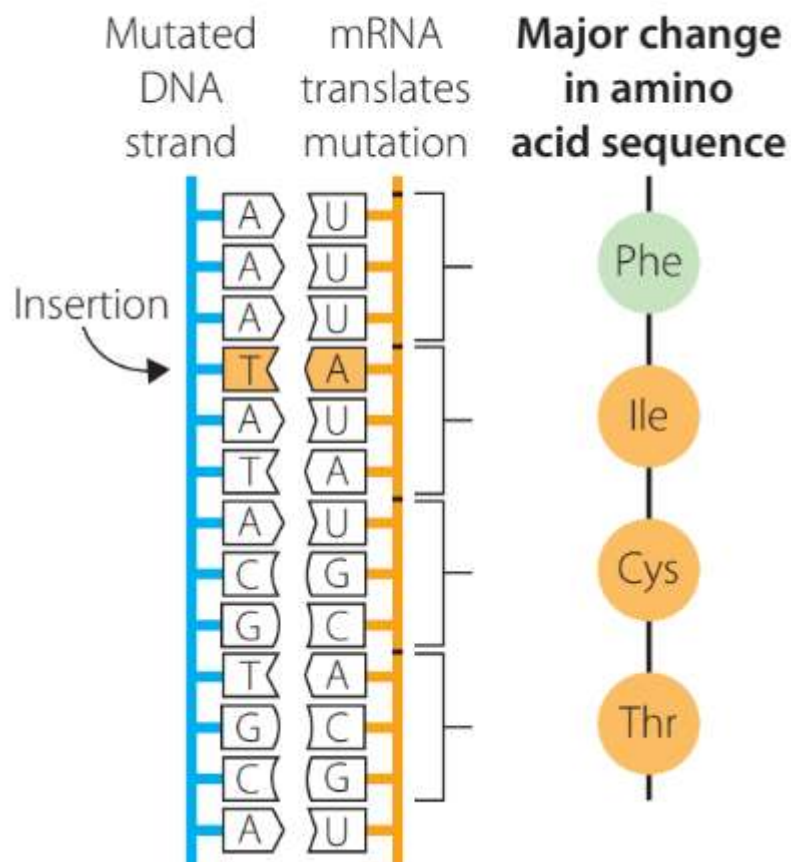
Frameshift Mutation:

- Happens when
 - 1 nucleotide
 - Inserted or deleted from dna
 - Change rna reading frame
- As mrna is read in codons
 - Change causes
 - All further codons to
 - Be misread
 - → completely diff seq.
 - Of amino acids
 - ∴ not working protein

- Happen from
 - Point mutations
 - Larger
 - Insertions
 - deletions
- harmful
 - If no. of inserted, deleted bases
 - Not multiple of 3
 - Making mrna read bases more wrong

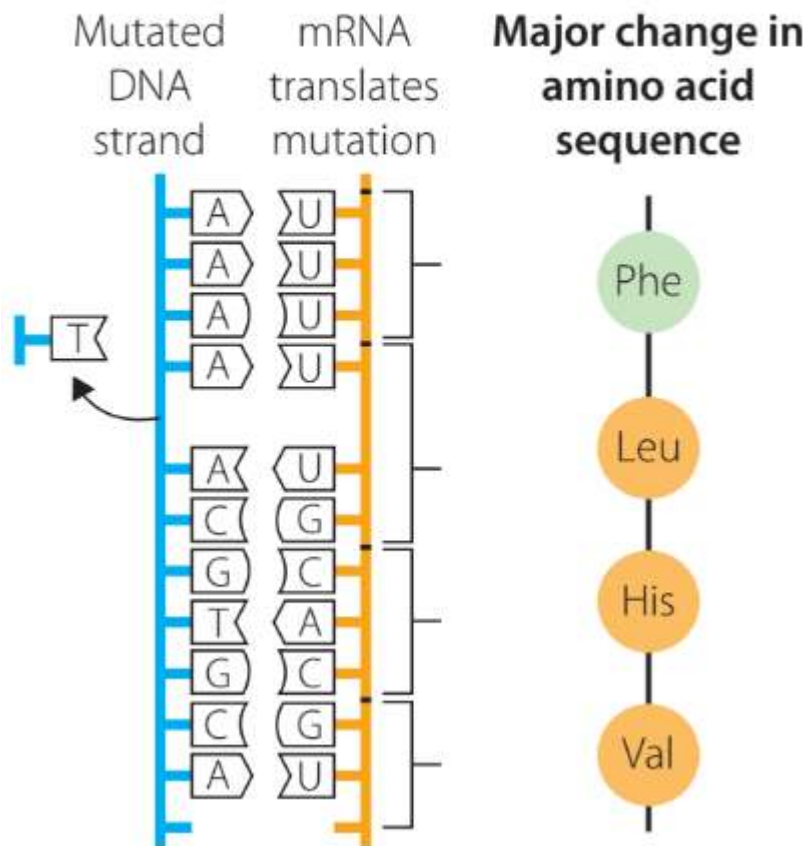
d

Frameshift insertion



e

Frameshift deletion



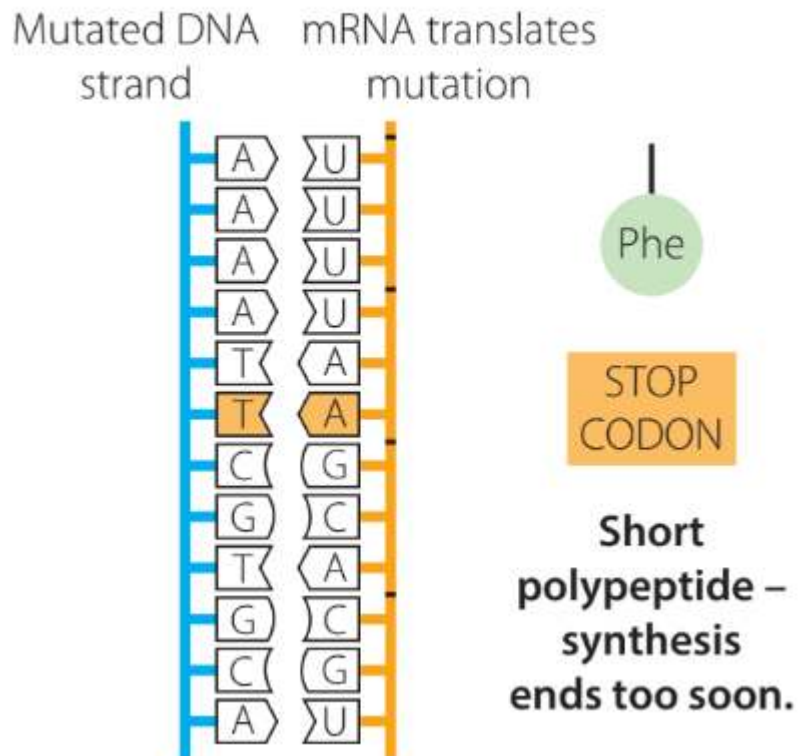
Changes in Proteins (Phenotype) due to Point Mutations:

Nonsense Mutation:

- Amino acid → stop codon
- Cuts protein short
- Result protein
- Non functional
- → big phenotype effect

a

Nonsense mutation

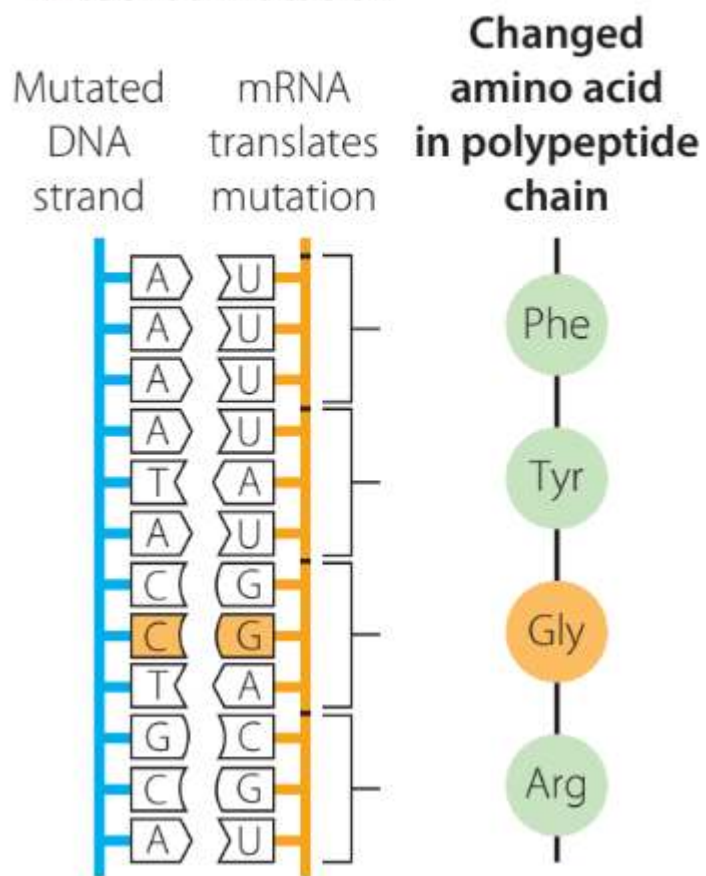


Missense Mutation:

- Result in
 - Amino acid change
- Functioning of result protein
 - Determined by
 - If replaced amino acid
 - Is same type
 - As original
- Protein end product changed
 - + less functional
- "Sense" or "meaning" of protein changed

b

Missense mutation

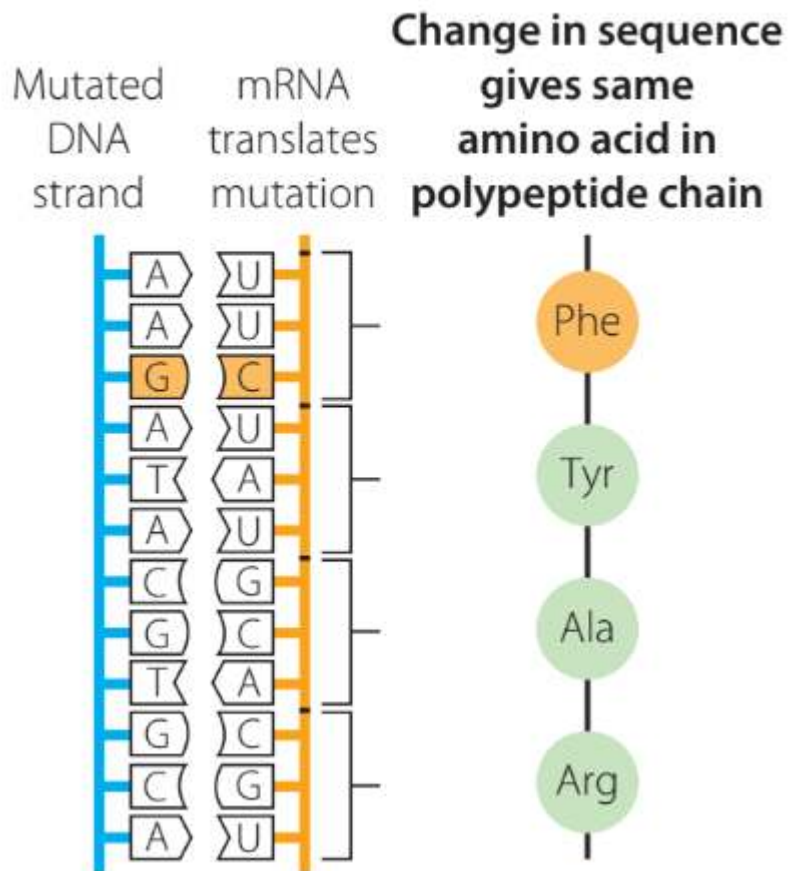


Silent Mutation:

- No change in amino acid
- Because of uselessness of genetic code
 - $\therefore$ no effect on proteins
- No effect on phenotype (protein)



Silent mutation



Neutral Mutation:

- Amino acid result
- Same type as original
- Change has no effect
 - On protein structure

Chromosomal Mutations:

- Affect large sections of
 - A chromosome
- Change in chromosome
 - Structure
 - Dna gene number
- Normally happen
 - During meiosis
 - By mutagens
- 4 types
 - Deletion
 - Insertion (duplication)

- Inversion
- Translocation

Deletion:

- Section of dna removed
 - Not replaced
- → reduction in no. of genes
 - In chromosome
- Happen by
 - Exposure to high heat
 - Virus
 - Radiation
- E.g. disorder:
 - Cri du chat syndrome

Insertion (duplication):

- Part of dna
 - Duplicated
 - Or x2
- + is inserted
- Increase no. of genes on
 - Chromosome
- Effect
 - Depends on
 - Size
 - Location
 - No. of repeats
 - If above a threshold (limit)
- Position of duplication
 - If its
 - Exon
 - Intron
- E.g. disorder:
 - Huntington disease

Inversion:

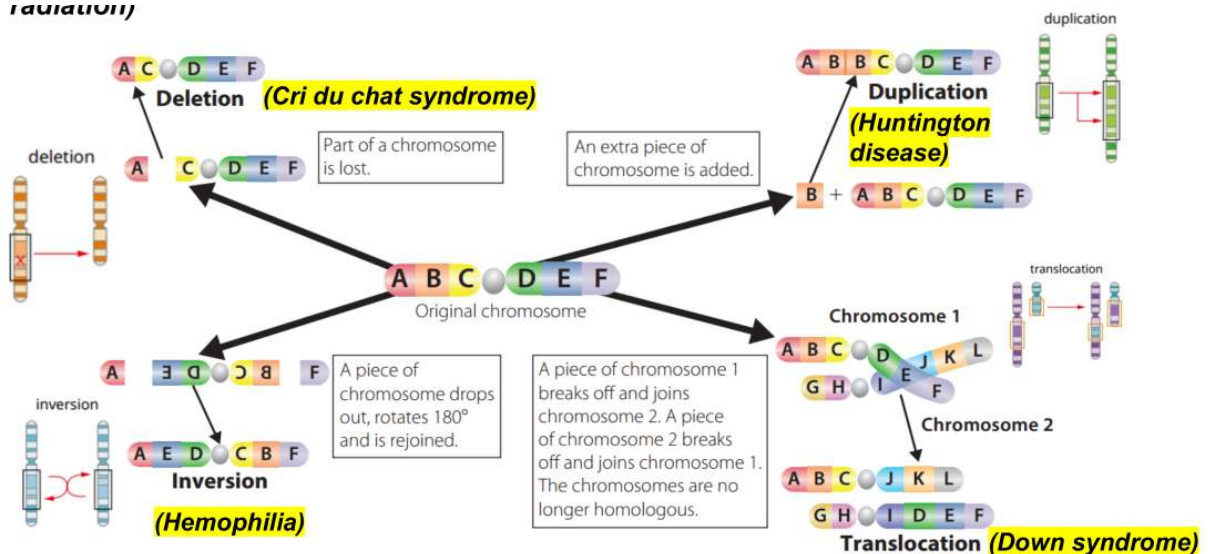
- Happens when

- Section of dna
- Removed
- + sequence reversed
- Then added into chromosome
- So bases are in
- reverse order
- E.g. disorder:
 - haemophilia

Translocation:

- Section of dna
- Moved from 1 chromosome
- → non homologous chromosome
- May → to
 - Gene fusion
 - Translocated region
 - Joins 2 normally separate genes
- E.g. disorder:
 - Down syndrome

radiation)



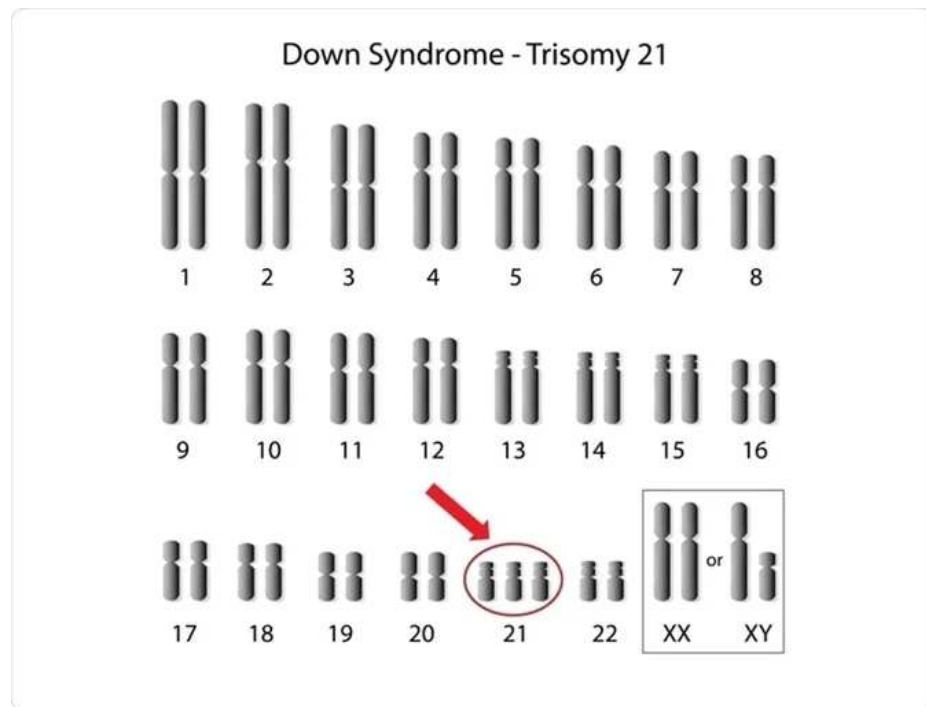
Chromosomal Abnormalities:

- Mutation involving
 - Whole chromosomes
 - No. of chromosomes

Aneuploidy:

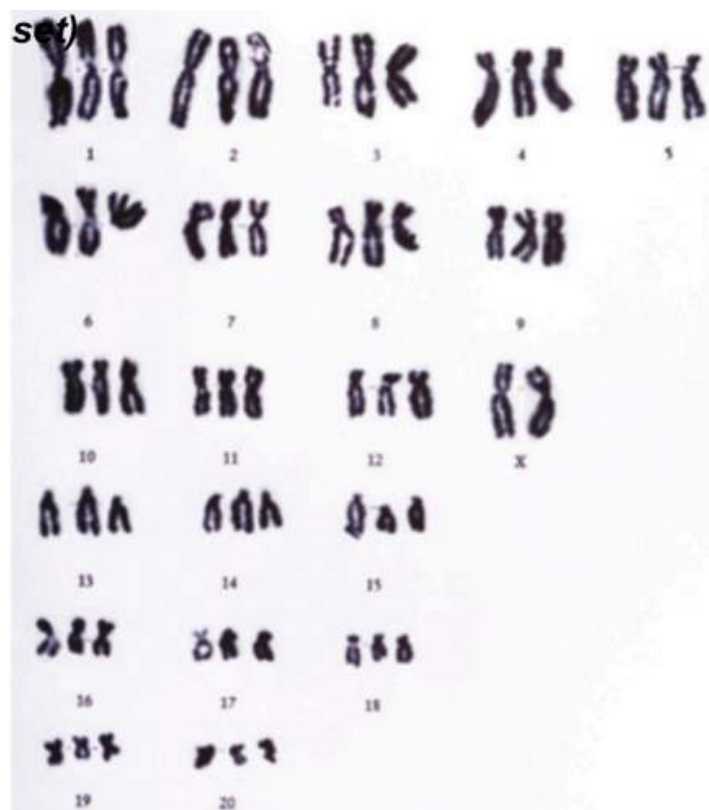
- Extra or missing 1 chromosome

- Down syndrome
- Extra copy of
- Chromosome 21
- (Trisomy)



Polyploidy:

- More than 2 in
- every homologous chromosome set



Mutations and How they Affect Organisms:

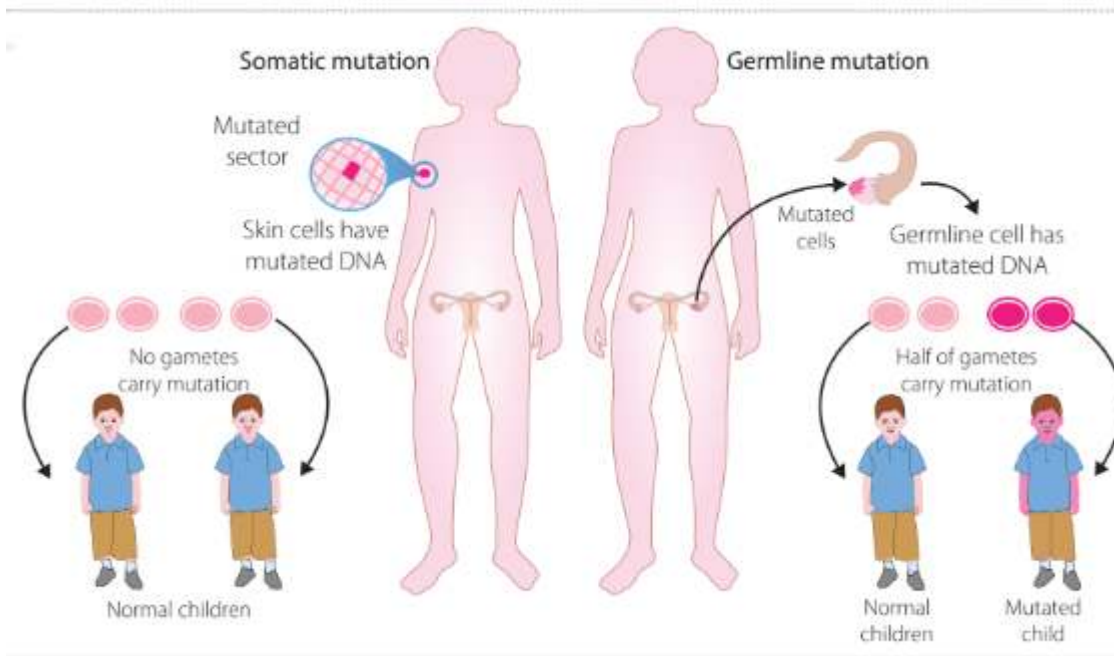
Somatic Mutations:

- In somatic cells
 - Body cells
 - Not reproductive cells
- Not inherited by offspring
- Happen by
 - Dna replication errors
 - In mitosis
- If mutation happen
 - Cell divide
 - Daughter cells get mutated dna
 - Make mutation situation bigger in tissue
- Result
 - Phenotypic difference
 - in tissue
 - E.g.
 - Pigmented cells (carcinoma)
 - Tumour
 - Disease observable in organ
 - Commonly cancer
- May cause physiological changes
 - In cells
 - Not always visible

Germline Mutations:

- (gametic mutations)
- In Sexual reproductive cells
 - Gametes
- Affects half of gametes
- Passed → offspring
- When gamete having mutation
 - Fuses with other gamete
 - → embryo
 - Mutation replicated

- In every cell of embryo
- $\therefore$ affecting all cells of child



Coding and Non-Coding DNA:

- Coding dna
 - Codes for proteins
- Non coding dna
 - Does not code for proteins
- Phenotype of individual
 - Depends on
 - Presence of particular proteins
 - Activity of those proteins

Coding DNA Mutations and their Significance:

- Affect
 - Type
 - seq. Of amino acids
 - In protein
- $\therefore$ affect
 - Structure
 - Function
 - Of protein
 - $\therefore$ affect
 - Phenotype of individual
- Prokaryotes

- Mostly coding dna
- Mainly genes for making
 - dna repair enzymes
- If inactivation of coding dna
 - Mutation Increased rate
- Prokaryotes + eukaryotes
 - Genes for dna repair enzyme making affected
 - Increase rate of mutation
 - From dna replication errors
- Mutations in
 - Tumour suppressor genes
 - Or proto oncogenes (promote cell division or reduce cell death)
 - → cancer

Non-Coding DNA Mutations and their Significance:

- Non coding regions have
 - Regulatory seq. That
 - Control if
 - Genes switched on
 - Genes switched off
 - + at what time
- Result in
 - Disruption of
 - Timing
 - Level of gene expression
 - ∴ affect
 - Cell function
 - + development
- Increase risk of disease
 - In adulthood
 - Cancer
 - Diabetes
 - Obesity
 - By altering gene regulation
- E.g.
 - Mutations in non coding dna regions
 - Linked to congenital abnormalities (birth defects)
 - Like heart defects

- Non coding dna may
 - Be a buffer
 - During genetic recombination
 - Or mutation events
 - ∴ reducing impact of
 - Harmful mutations

Causes of Genetic Variation:

- Mutation
 - Increases no. of alleles
 - For a trait

Meiosis and Genetic Variation:

- Crossing over
- Random segregation
- Independent assortment
- Result in
 - Recombination of
 - Maternal
 - + paternal genes
 - In each gamete
- Fertilisation
 - Further mixing of genetic material
 - Result
 - Offspring with many new
 - Combinations of genes
- Variation during meiosis
 - Caused by mutations
 - During
 - Dna replication
 - Separation of chromosomes (disjunction)
 - → chromosomal mutations
 - Replication errors in meiosis
 - Happen in similar manner
 - To mitosis
 - → point mutations

Chromosomal Errors:

- From
 - Crossing over errors
 - Non-disjunction of chromosomes
- E.g.
 - Dna that is exchanged during crossing over
 - Is inverted before
 - being inserted onto
 - Arm of matching chromatid
 - Chromosome may break
 - + if followed by duplication of chromosome
 - The 2 Chromatids that fuse
 - Have 2 centromeres
 - When pulled apart in anaphase
 - 1 chromatid has duplication
 - 1 has deletion

- Chromosomal changes

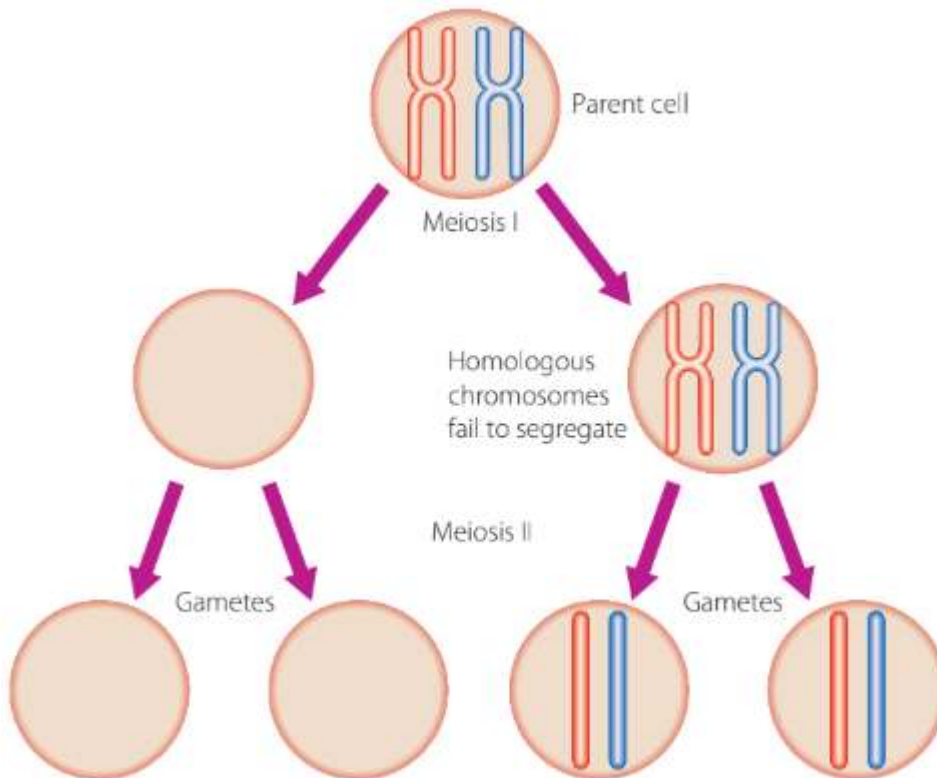
- May happen due to
 - Exposure to mutagens
 - during meiosis
- Female exposure to mutagens when
 - Their eggs are still in meiosis 1
 - As they have eggs
 - from embryonic life

Changes in Chromosomal Numbers (Non-Disjunction)

- When 1+ pairs of chromosomes

- Or sister chromatids
 - Don't separate as they should
 - When dividing nucleus
- Abnormal distribution of
 - Daughter chromosomes
 - In resulting cell
 - → change in chromosome no.
- Increased risk of errors from
 - Increased age of parents
 - Some environmental factors

- E.g.
 - Down syndrome
 - Extra copy of chromosome 21



Mutation, Gene Flow and Genetic Drift:

Genetic Diversity:

- From
 - Mutations
 - Migration (emigration, immigration)
 - Presence of multiple alleles
 - In a population
- Involves many variants (alleles)
 - For each gene
 - + many genes for each trait
 - Esp polygenic traits
 - E.g. height
- Polygenic traits
 - Wide range of
 - genotypes + phenotypes
 - Due to
 - multiple interacting genes

- Mutations
 - are
 - Key source of
 - Genetic variation
 - Can affect
 - Phenotypes
 - Influence
 - Natural selection
- Most mutations are
 - Neutral
 - Harmful
- Beneficial mutations
 - Rare
 - But can spread through population
- Neutral mutations
 - May be kept
 - As evolutionary backup
 - Provide
 - Potential advantages
 - If environmental conditions change

Factors that Cause Changes in Allele Frequency within a Population:

Natural Selection (Selective Pressure):

- Alleles that
 - Improve individuals ability to
 - Survive + reproduce
 - Are more likely to be
 - Passed on
- Over gens
 - Favourable alleles increase
 - Unfavourable alleles decrease
- Above point drives
 - Adaptive evolution
 - As populations
 - Become better suited to
 - Their environment

Sexual Selection:

- A form of
 - non random mating
- Where individuals
 - With desirable traits
 - Bright feathers
 - Strong build
 - Are more likely to reproduce
- Increases
 - Freq. of alleles
 - Linked to mating success
 - Not necessarily survival

Mutation:

- Gets
 - new alleles
 - into gene pool
 - By causing changes in dna
 - During meiosis
- Changes may be
 - Beneficial
 - Neutral
 - Harmful
- Only mutations in germline cells
 - Heritable
 - Can affect population level allele freq.

Genetic Drift:

- Random changes in
 - Allele freq
 - Due to
 - Chance
 - Not selection
- More pronounced in
 - Small populations
- Includes
 - Bottleneck effect:
 - Dramatic reduction in population size

- E.g. natural disaster
 - Randomly alters allele frequencies
- Founder effect:
 - Small group of individuals
 - becomes isolated
 - from main population
 - Allele freqs may differ by chance
 - Leading to new population
 - That is genetically distinct
 - + reduced genetic diversity

Gene Flow:

- Movement of alleles between populations

- Due to
 - Immigration
 - Emigration
- Introduces new genetic material
 - Increase variation
- Alleles may also be lost
 - From population due to
 - Individuals leaving
- Has stronger impact on
 - Smaller populations

FACTOR	WHAT IS IT	CHANGE IN ALLELES IS DUE TO ...	EFFECT ON THE NEXT GENERATION
Selective pressure	The main selective pressure is natural selection (Darwin)	variations that are passed on because they make individuals more likely to survive (and more virile)	Alleles that make individuals 'fitter' – more likely to survive and live to reproductive age – become most frequent
Sexual selection	Certain individuals are more attractive to mates and therefore more likely to breed	non-random mating (mating is not random; some individuals mate more than others)	Alleles of individuals who are most successful at mating are more common in the gene pool
Mutation	New genes arise due to 'errors' in DNA replication during meiosis (gametogenesis); they may be beneficial, neutral or harmful	new alleles arising during gametogenesis being introduced into a population	New alleles that are beneficial become more frequent in the population
Genetic drift (more obvious in smaller populations)	Random events (e.g. a tornado) lead to a change in gene frequency because some individuals are wiped out	random chance (non-selective; does not depend on genetic make-up)	Causes individuals within a population to be different (not necessarily more successful) due to random chance
Gene flow (more obvious in smaller populations)	Individuals with different genes come into a population and spread their alleles	mixing with new genetically different individuals (e.g. immigration, emigration)	Allele frequency in the population changes

Biotechnology:

- Use of biological materials (bio)
 - As tools (technology)
- Use of living organisms
 - Or their products to
 - Fulfill human needs
- Idea that
 - living materials
 - biological processes
 - Biological products
 - Are used to make new products
 - Useful to humans
- Fields:
 - Industry
 - Agriculture
 - Medicine
- Examples:
 - Bio tech materials from living things
 - Cell products
 - Enzymes
 - Antibiotics
 - Living cells
 - Tissues
 - Whole organisms
 - Micro organisms
 - Plants
 - Animals

Past, Present and Future of Biotechnology:

- 1000s years old practices
 - Ancient biotech
- Late 1800s → 1950s
 - Classical biotech
- 1950s → present
 - Modern biotech
 - From discovery of dna → present

Past biotechnology:

- In early times
 - Biotech developments based on
 - Using products available in nature to improve
 - Living conditions + survival
 - Targets basic human need for
 - Food
 - Shelter
 - Clothing
- Ancient biotech in practice
 - Without full understanding of
 - biological + chemical processes involved
- Practices included:
 - Use of yeast to make bread
 - Beer + wine fermentation
- Yogurt + cheese
 - made using bacterial cultures
- Classical biotech:
 - Pasteur
 - Mendel
 - Penicillin

Present biotechnology:

- Discovery of dna
- Dna manipulation at molecular level
- Gene tech
 - Manipulation of dna → make products
 - for human use
- Genetic engineering
 - To make specific proteins
 - Regulate cell processes
 - So specific outcomes achieved
 - + precise needs met
- Fields:
 - Industry
 - Agriculture
 - Medicine

- Gm (genetically modified)
 - Food
 - Organisms
- Making industrial products
 - Paper

Future biotechnology:

- Advances in genomics
- + understanding of how proteins are expressed
 - Used for improving treatment of
 - infectious diseases
 - Cancer
 - Other genetic disorders

Ancient Biotechnology:

Agriculture:

- Humans worked to improve
 - Quality + yield of their food
 - By selecting seeds from best crops
 - + breeding best quality animals
- Was start of selective breeding or artificial selection
 - Involving human 'manipulation'
 - of living organisms

Food Production - Bread and Cheese Making:

Bread Making:

- Process of bread making
 - Shown in carvings in
 - Walls of tombs in ancient egypt
- Explains need for wheat from
 - seeds that could be naturally extracted
- Yeast in bread accidental
 - After was intentionally put in
- Humans from migratory species → settled groups
 - Learned how to use plants and animals around them
- Maize, beans, potatoes
- Rice, millet, buffalo

- Wheat, sheep, goats

Cheese Making:

- Involves action of
 - Bacteria on milk
- 10 000 years ago
- Process improved over years
- Adding enzymes
- Curds and whey produced
- Pressing curds
 - Removes excess water
- Storage in dry environment
 - Allows ripening
 - Action of surface bacteria to give cheese flavour

Classical Biotechnology (1800s to 1950s):

- Uses
 - Known biological material such as cells
 - Yeast
 - + Other microbes
 - + known cell products
 - Enzymes
 - + antibiotics
- To achieve particular goal

Fermentation:

- First 6000 years ago
- Ancient Egypt
- Louis Pasteur
 - Discovery of process from yeast
 - From gas bubbles in beer
 - Experiment to show fermentation
 - As biological process

Medicine and Antibiotic Production:

- Conventional drugs + medicines
 - Derivatives of ancient remedies
- Awareness of medicinal plants

- from ancient times → today
 - Led modern scientists creating medicine
 - from plants and fungi
 - Which is used in modern pharmaceutical drug development
- Penicillin:
 - Product made by fungus penicillium
 - 1928
 - Alexander Fleming
 - Was later stabilised by Australian pharmacologist + pathologist
 - Howard Florey
 - To make penicillin

Plant Selective Breeding:

- Hybridisation
 - Produces hybrid offspring with combined traits
 - Make them better suited
 - to their environment
 - Or their end use
- Artificial pollination
- Traits good for breeder or consumer
 - May be bad for plant
- Traits that increase a plant's resistance to disease
 - May be lost in turn for enhancing other traits

Modern Biotechnology (1950s to Present):

- Genetic engineering:
 - Techniques used to manipulate DNA
 - To meet the needs of
 - a society
 - The environment
 - And science
 - In a very precise way
 - Involves
 - Humans manipulating pattern DNA bases
 - (genotype)
 - To change its phenotype
 - Making organism appear or act differently
 - Common steps

- Cutting, copying + pasting dna sequences
- Example:
 - Human gene snip cut using enzymes
 - Gene → bacteria
 - Bacteria makes human proteins
 - Insulin
 - Human growth hormone
- Dna made from one or more species
 - Recombinant dna (rDNA)
- Organisms with rdna
 - Genetically modified organisms (gmos)
 - If they can pass new dna → offspring
 - Are transgenic species
- Dna technology
 - Biological tools to modify measure manipulate + manufacture dna
 - Many of the tools are enzymes
 - From other living organisms
 - Bacteria
 - Viruses

Applications of Modern Biotechnology and Social Implications:

Industrial Biotechnology Applications:

- Includes:
 - Enzyme engineering
 - Bio nanotechnology
 - Synthetic biology
 - Bio material engineering
- Applications:
 - Pollution prevention:
 - Microorganisms
 - clean up + reduce waste
 - Enzymes in washing powders
 - to remove stains
 - Replacing phosphates that were previously used
 - + built up in water systems
 - Making environmentally friendly products
 - Fertilisers

- Biopesticides
- Biofuels
- Biodegradable materials
- Made from plants
- Making biomaterials
 - Involves any natural or synthetic substance being made to
 - Interact with biological systems
 - For
 - Medical
 - Therapeutic
 - Diagnostic purpose
- Future applications:
 - Biofabrication
 - Automated production of tissues and organs
 - Using principles of 3d printing
 - Materials like
 - Cells, fibres + gels
 - To replace diseased or damaged tissue
 - Synthetic biology:
 - Digital tech used to make synthetic genomes
 - That can function in living cell

Agricultural Biotechnology Applications:

- Improving plant + animal production by
 - Increasing yield
 - Nutritional value
 - Resistance to disease
- Agric biotech
 - Enhance quality of
 - And economic returns from
 - Commercial crops by
 - using reproductive technology
 - + gene technology

Reproductive Technologies and Genetic Diversity:

- Artificial insemination
 - (animals)
- Artificial pollination:

- (plants)
- In vitro fertilization
- Embryo transfer
- Cloning
- Advantages of selective breeding in agriculture include
 - Increased
 - Milk yield (female cattle)
 - Quality of beef in cattle
 - Size of eggs from chickens
 - + more frequent laying
 - Grain yield wheat
 - Food protein
 - Food vitamin content
 - E.g. golden rice
- Genetically modified crops
 - Golden rice
 - Used to address hunger in poor, developing countries
 - + work towards eradicating starvation
 - Created using
 - Insertion of 2 genes
 - 1 from daffodil
 - 1 from bacterium
 - Into a rice genome
 - Has vitamin A precursor
 - Beta carotene
 - Used for humanitarian causes
 - Combat vitamin A deficiency in developing countries
 - Deficiency causes
 - Blindness
 - Other health problems
 - Concern about
 - Loss of genetic diversity
 - + biological resources
 - Due to farming generally +
 - Farming in developing countries

Conservation Biology Applications:

- Plant biotech for

- Biodiversity conservation
- Advances in genomics and proteomics
 - allowed researchers to
 - Identify desirable traits in crops and plants
 - Using molecular markers
 - Save good germplasm
 - Germplasm:
 - Any living tissue from which new plants can form
 - Includes whole plant parts
 - Leaves, stems
 - Pollen, clusters of cells
- Animal biotech
 - Include researchers mapping relatedness
 - between animals (in wild vs in captivity)
 - Then designing breeding programs to
 - Maximise
 - hybrid vigour
 - Genetic biodiversity
 - Artificial insemination for animal conservation

Medical Technology Applications:

- Delivering normal genes that function correctly
 - To people who don't have functional gene
- In vitro fertilisation (ivf)
 - Making zygote in test tube
 - Instead of in female
 - Allows couples with fertility issues
 - To reproduce
 - Prevents children being born with
 - Inherited genetic diseases

Ethics and Social Implications:

- Ethics:
 - Moral principles, standards, values
 - Concerned with what is
 - Good
 - Right
 - Just

- Reasoned views behind these judgements
- Social implications:
 - Effects + consequences that
 - Actions, technologies
 - Have on people and society
- Influencing
 - how they interact
 - Their communities and overall well being

Making an Ethical Decision:

- Bioethics:
 - Study + investigation of
 - How decisions in medicine + science
 - Affect society + environment

TABLE 8.1 Basic principles of ethics

	PRINCIPLE	QUESTIONS	AIM
1	Benefit and non-harm	In whose best interests is the action – does it benefit individuals/groups/society in general/future generations? How reliable and useful is the procedure? Do the advantages outweigh the risks?	Wellbeing
2	Individual rights and autonomy	Is the person properly informed of the advantages and harms or risks? Does the person have freedom of choice? Are the rights of others protected or breached?	Choice
3	Privacy and societal perception	Will the person's genetic information remain private? What is the right thing to do if a breach of privacy could save another person's life? How might the release of this information affect the way in which society or another individual perceives that person?	Respect
4	Equity and justice	Is there equal access for all members of society (no bias in terms of gender, socioeconomic status or ethnicity)? Is there fair sharing of resources that can be justified? How can discrimination be avoided?	Fairness

Specific Biotechnologies Ethics and Social Implications:

Selective Breeding, Artificial Pollination and Hybridization:

- Selective breeding
- Artificial pollination
- Hybridization
- All used to get
 - desired combination of favourable characteristics
 - In organism
- Plant e.g.
 - Corn bred selectively
 - To have bigger
 - + more kernels
- Animal e.g.

- Labrador, poodle
- Characteristic for family friendly dog
 - Hypoallergenic

Ethical Implications:

- Exaggerated features
 - May negatively affect health of animal
- Debate on if
 - Increase in global food and increased nutritional value
 - Is used to support developing countries
 - Or make an obese population more obese

Social Implications:

- Increase in yield
- Less land needed from
 - Pesticides
 - Size
 - Resistance
 - Faster growth
- Pest resistance can reduce pesticide residue
 - That may harm surrounding environment
- Increase in health benefits of food
- Higher profit
- Faster reproduction
 - So endangered species encouraged
- Less environmental damage
 - Due to
 - More efficient food production
- Extinction susceptibility
- Inbreeding → diseases
- Reduces biodiversity
 - Due to constantly selecting same traits

Vaccines, Medicine and Antibiotics:

- Vaccines:
 - Weak, inactive antigens
 - That trigger immune system
 - To develop resistance to it

- For future infections
- Medicine + Antibiotics:
 - Medicinal plants used to make medicine
 - E.g. penicillin:
 - Found to be produced by fungus
 - Stops bacteria from growing

Ethical Implications:

- Developing nations who need these
 - Most can't afford
- Vaccines + antibiotic tested on animals
 - Can harm health
 - Cause pain or distress
- Intentionally breeding animals for research
 - May harm them

Social Implications:

- Less burden on health care systems
 - As people are healthier through
 - Use of plants for medical treatment
- Mould penicillin has antibacterial properties
 - Found to treat infections
- Prevent disease before outbreak happens
- Eradicate disease
- Longer life span
- Resistant strains of pathogens
 - From inappropriate use of antibiotic or antiviral
- Changing genomes of other organisms
 - + introducing them → ecosystem
 - Lead to
 - Invasive behaviour
 - + reduce biodiversity
- Finding plant or pharmaceutical purposes
 - Raise value of material
 - Result in
 - Exploitation
 - → species loss

Transgenic Species:

- 1+ dna sequences from another species
 - Introduced using artificial means
- Animals made transgenic by
 - Having small sequence of foreign dna
 - Injected → fertilised egg, developing embryo

Ethical Implications:

- Unfair sharing of benefits
 - Depend on who has access to genetic resources
- May cause organism to have features that harm them
- Genetic engineering + selective breeding
 - Appear to violate animal rights
 - By manipulating them for human needs
 - As if they were nothing more than human property
 - > treating animals as being of value
- Vegetarians, religious people
 - Concerned with animal genes
 - When used in plant products
- Is adding human genes to animals immoral?
- Negative environmental impact

Social Implications:

- Can manipulate food to
 - Have greater nutritional value
 - (addressing hunger in poor and developing countries
 - Golden rice - high vit a)
- Increase yield
 - Disease +
 - Pest resistance
 - Weather resistance
 - Enhanced flavour
- No need for chemical pesticides
 - Less damage to environment
- Potential pharmaceutical uses
 - E.g. tracy the sheep
 - First transgenic mammal created
 - Produced human protein in milk

- That was used as potential pharmaceutical
 - For treatments of cystic fibrosis + emphysema
- Transgenic organisms become dominant
 - as they were modified to increase survival
 - $\therefore$ causing extinction of natural organisms
- Organisms $\rightarrow$ ecosystem
 - Lead to invasive behaviour
 - $\rightarrow$ loss of biodiversity
- We don't know long term effects of eating gmo foods
- Farmers in developing workforces
 - Cant compete in markets
 - As they don't have access to gmo technology
 - Social inequality

Future Directions:

Cloning:

Somatic Cell Nuclear Transfer

- Remove mature somatic cell (body cell)
 - Like skin cell from animal
- Transfer dna of donors somatic cell
 - $\rightarrow$ egg cell
 - Whose dna/nucleus removed

Implications:

- Can increase genetically 'perfect' animals or crops
 - That can provide greatest produce
 - Output
 - Nutrition
 - Quality
 - Growth rate
- Helps with regenerative medicine + stem cell treatment
 - For
 - Burn victims
 - Reconstructive surgeries
- May help plants, animals who cant reproduce
- Source for organ replication
 - For organ transplants

- reduce genetic diversity
- Cloning an organism not provide
 - Exact replication as
 - Environmental factors affect phenotype expression
- If one organism has disease, fault or dies from disease
 - All other organisms will be affected accordingly
- Cloning against certain religions
- uncertain/unpredictable results:
 - Over 89% of animals and human attempts have failed
 - Animal dna only exposed to risk
- Only rich can afford

Gene Editing / Splicing:

- Cutting out genes
 - using restriction enzymes
 - Cas9
 - Cuts dna at a specific base sequence or gene
- CRISPR-Cas9
 - Enzyme
 - Found in bacteria
 - Used to cut dna at psp
 - Enzyme contains guide DNA
 - Complementary base sequence to targeted gene
 - Cas9 dna cutting enzyme
 - Introduced to cell
 - Locates dna strand + cuts it
 - Replacement healthy dna strand replaced

Implications:

- May completely treat disease quickly
 - Not just symptoms of it
 - Reduce + eliminate future health consequences
 - By removing harmful dna mutations
- Some gene editing tools inexpensive
 - Compared to current therapies
 - ∴ benefit individuals + economy
 - CRISPR
- Is recent technology

- Not fully researched
- Potential side effects
 - Maybe cancer
- May reduce biodiversity (GM mosquitoes)
- Gene drive
 - Help improve “organism”
- Being able to edit + design babies
 - Controversial topic
- Future inequality in society
 - “Deisgner babies”
 - Have enhanced physical features,
 - Intelligence
 - Athleticism
 - Which may create superiority in society
- Against religious beliefs
 - May cause racial discrimination

Benefits of Research of Genetic Technologies:

Genetic Technologies:

- GMOs
- Gene editing
 - CRISPR-CAS9
- Cloning

Benefits:

- Detailed implications, benefits above
-

- Improving human health:

- Gene therapies
- Gene editing
- Disease treatment
 - From genetic engineering
 - + cost effective, large scale production
 - Of essential drugs and hormones
- Disease prevention
 - Used to produce vaccines
- Engineering resistance in organisms

- Diagnostic + precision medicine
 - Identify individuals at risk for serious disease
 - → earlier diagnosis
 - Personalised treatment
- enhance
 - crop yields
 - nutritional value
 - More resistance to pests diseases herbicides
- Less environmental impact
 - Pest resistant crops
 - No need for harmful chemical pesticides
- Meeting growing demand:
 - Improved agricultural productivity
 - → can feed growing population

Changes to Earth's Biodiversity:

- Short term
 - Genetic technologies like crispr
 - Increase biodiversity
 - By introducing new gene combinations
 - In population
 - + new genes in individuals
- Long term
 - Breeding individuals with same desired features
 - Over generations
 - Decrease biodiversity as
 - All individuals genes
 - will be same or very similar
 - Use of reproductive technologies like
 - Cloning
 - Selective breeding
 - Increase biodiversity
 - In farming practices
 - Decrease biodiversity
 - Technologies like
 - Selective breeding
 - Cloning

- Genetic engineering
- Long term
- ∴ Increase risk of populations being wiped out
- In response to
 - Disease
 - Sudden environmental change
- Extinction of natural species
 - From gmos outsurviving natural population
 - If genes are better than populations genes

Genetic Technologies:

Uses and Advantages of Current Genetic technologies:

- Use gene editing + gene therapy

Selective Breeding:

- Breeding one organism with 1+ desired feature
- With other organism with 1+ desired feature
- Result offspring has 1+ desirable trait
 - From both parents
- Induce genetic change indirectly

Uses:

- Producing organisms with desired traits
 - From 2 separate organisms
- High crop yields
- Livestock productivity
- Increased disease resistance in plants, animals
- Plants, animals with specific aesthetics

Advantages:

- High crop yields
- Livestock productivity
- Increased disease resistance in plants, animals
- Plants, animals with specific aesthetics
- Economic benefits
- Increase food production
- Improved health and welfare

Gene Therapy:

- Taking desired gene
- Inserting it into modified viruses
 - That can't cause disease
- Virus delivers genetic material → cells nucleus
- Cause cell to express gene

Uses:

- Repair or replace faulty genes
 - In attempt to cure disease
 - Improve body's ability to fight disease
- Done at embryonic stage
 - To prevent heredity disease development
- Regulate gene expression

Advantages:

- No donor or other organisms dna needed
- Promising for
 - Treating wide range of diseases
 - Cancer
 - Heart disease
 - Diabetes
 - AIDs

Gene Editing - CRISPR:

- Cutting + removal of gene
- Using cas6 enzyme to cut
 - At particular base

Uses:

- Tool for gene therapy
- Edit dna sequence by cutting out
 - Or putting in nucleotide bases
- May develop new plant varieties
 - Which are indistinguishable from those
 - Developed from traditional breeding methods
- Gm mosquitoes
 - Targeted gene for fertility in males

- Reduce reservoir size by
 - Reducing offspring by adding gene to kill them

Advantages:

- May correct genetic defects
 - Treating
 - Preventing disease

Processes and Outcomes of Reproductive Technologies:

Artificial Insemination:

Processes:

- In animals
 - Collecting sperm from chosen male
 - Introducing into several selected females
 - Sperm collected from male
 - Collected semen is
 - Divided into semen straws
 - Chilled
 - Then frozen in liquid nitrogen
 - For long term storage
 - + transportation
 - When transferring semen → female
 - Semen straw thawed
 - Put into sterile insemination gun
 - Gun inserted → vagina
 - Semen deposited

Outcomes:

- Frozen sperm
 - Overcomes problem of
 - Transporting large animals over long distances
 - Cost effective
 - Reduces danger of animals of
 - Injury during
 - Transit
 - Mating
 - Many females may be inseminated

- 1 male can sire offspring with several females
- Semen frozen indefinitely
 - Male can still produce offspring
 - Many years after animal has died
- Insemination
 - Used in conservation
 - To increase numbers of endangered species
- May be costly
 - Specialised equipment
 - Time consuming
- May cause injury to female
 - If carried out incorrectly
- Reduced genetic diversity in populations
- Advantages far outweigh disadvantages
- Altered composition of breed population
 - Controlled breeding of selected population
 - Selective breeding of individuals with desired alleles
 - $\therefore$ alleles selected for are chosen by breeders
 - Instead of nature

Artificial Pollination:

- Selective breeding in plants
- Used in ancient times

Process:

- Removing stamens (male part) of a flower
- Dusting pollen onto
 - Stigma (female part) of same flower
 - (self pollination)
 - Or different plant flower
 - (cross pollination)
- Breeder has more control over breeding process
- Hand pollination
 - Horticultural technique
 - Allows production of offspring with
 - Specific favourable characteristics
 - E.g.
 - Disease resistant fruit

Outcome:

- Allows for creation of new varieties of plants
- Overuse
 - → to crops that are too similar
 - ∴ reduce biodiversity
- INsect pollinated gooseberry plants
 - Had larger fruit
 - Greater seed germination rates
 - Than artificially pollinated plants
- Produce hybrid plants
- Increase genetic variability in populations
 - Due to creation of hybrid species
- New alleles introduced to
 - Gene pool of population
- New gene combinations passed to
 - Future gens if
 - Hybrids fertile
- Hybridisation in 1 species
 - Fertile offspring
- Hybridisation across species
 - Infertile offspring

In Vitro Fertilisation:

Process:

- Egg is fertilised by sperm
 - Outside of mothers body
 - In artificially created environment
- Resulting zygotes cultured
 - → early stage of development
- Zygote → mother
- Or zygote frozen in liquid nitrogen
 - For research
- Embryo → uterus
 - Through catheter (thin tube)
 - → uterine lining
- Used when
 - decreased fertility in

- 1 or both parents

Outcomes:

- Genetic diversity reduced
 - In population
 - Due to making of large numbers of embryos
 - From small selection of parent animals with
 - Desirable traits
- Genes for infertility
 - Now inherited by offspring
 - Otherwise would have not been inherited
- Sperm banks
 - Potentially alter genetic composition of population
 - In animals + humans
 - People may select sperm donor traits
 - Increase freq. of specific same traits
 - Decrease freq. of other traits

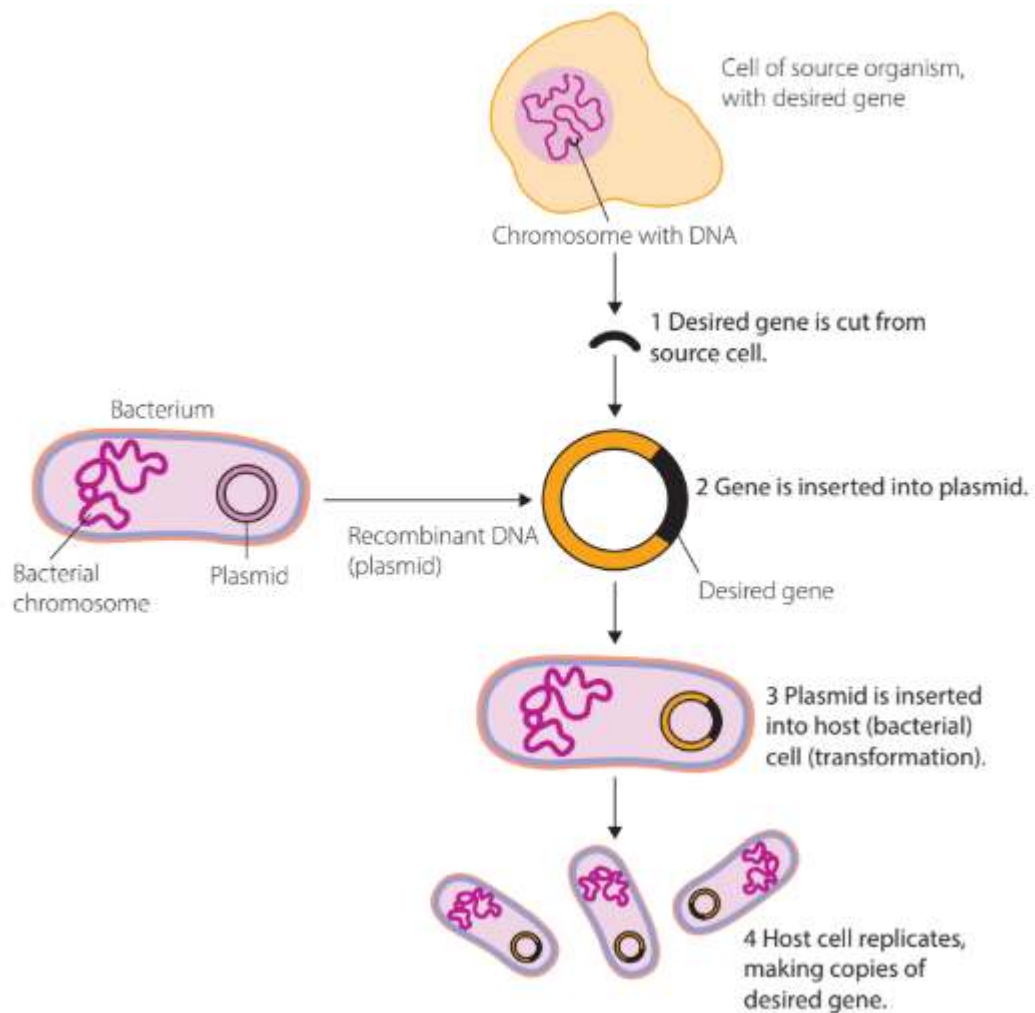
Cloning:

- Making of exact copy
 - Of organism
- Gene cloning
 - Cellular level
 - Produces identical copies of one gene
 - Multiple copies of a gene
 - Needed for genetic engineering
 - + biotechnological research
- Whole organism cloning:
 - Aka reproductive cloning
 - Create genetically identical
 - Whole organism using
 - 1 or more somatic cells
 - From another mature organism
 - Form of asexual reproduction
 - A reproductive technology

Gene Cloning:

Through Bacteria:

- Process
 - Select gene
 - Remove gene from source dna
 - Insert into dna of another organism
 - Make identical copies of that gene
- Technique used in insulin production
 - On a large scale
- Simple steps
 1. Gene (section of dna)
 - Cut from source organism
 - Using restriction enzymes
 - (enzymes made by bacteria)
 2. Gene pasted → vector dna or plasmid
 - Use ligase enzyme
 3. Plasmid having gene → host cell
 4. Host cell now make copies of vector dna
 - When making own dna copies



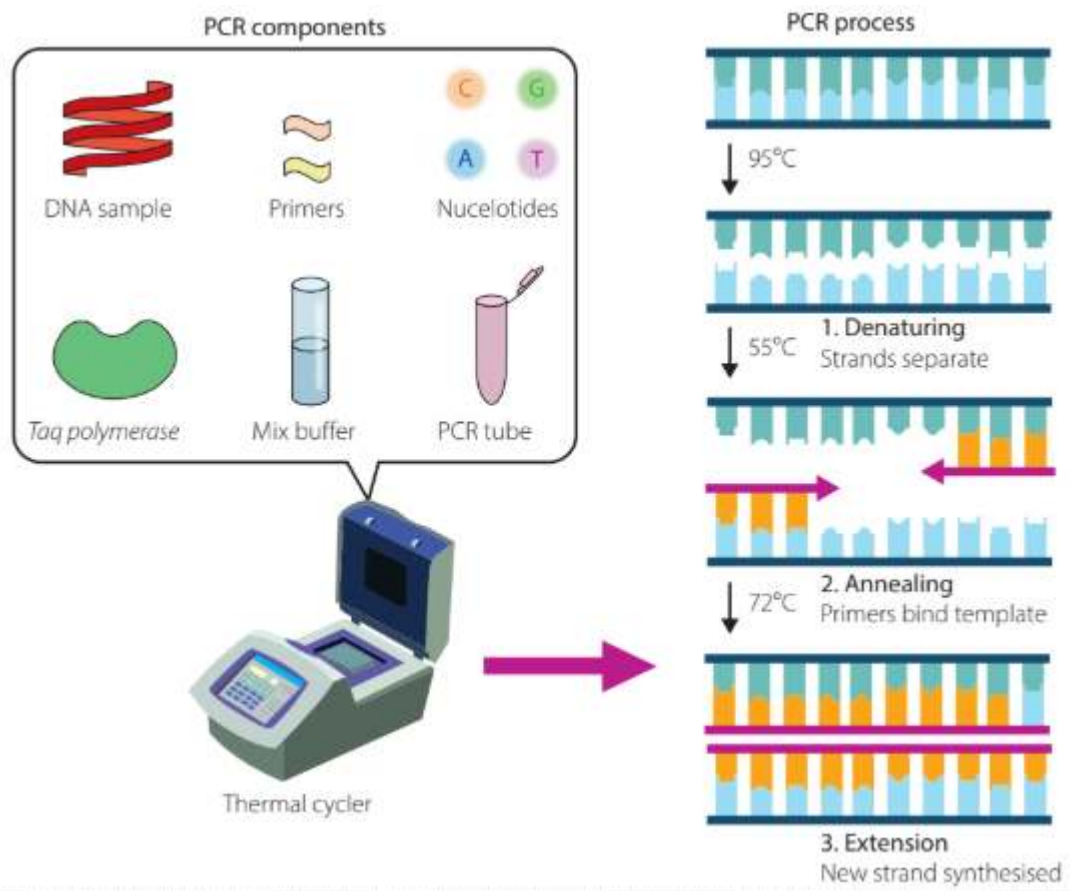
Polymerase Chain Reaction (PCR):

- Form of **in vitro dna cloning**
- **Carried out in test tube**
 - Instead of living organism
- Used in research
- Amplifies dna sequence
 - Amplifies = makes many copies of
- Process of
 - **thermal cycling to denature dna strand**
 - **Use of complementary primers**
 - **Locate + duplicate section of dna**
- Detailed steps:
 1. denaturation
 - Test tube heated → 96 degrees c
 - Break hydrogen bonds between nucleotides
 - Result is 2 single stranded halves of dna
 2. annealing

- Test tube cooled → between 55 and 65 degrees c
- Bonds form
- 2 dna primers used
- Forward primer → 3' end
- Reverse primer → 5' end other strand

3. extension:

- Test tube heated → 72 degrees c
- Specialised dna polymerase
 - Joins free floating nucleotides
 - With template strands
- Whole cycle repeated 25-35 times
- Produce millions of copies of dna strand
- Use heat stable dna polymerase
 - As normal polymerase denatures
 - When increase heat



Effectiveness:

- Allows copying of genes
 - That can be used in combination
 - With other genetic technologies

- To help us
- Quick copying of genes
- Inexpensive
- produces large amounts of specific genes
- Good for developing gmos
 - + advancing medical treatments + agriculture
- Targets + replicates desired genetic material

Whole Organism Cloning:

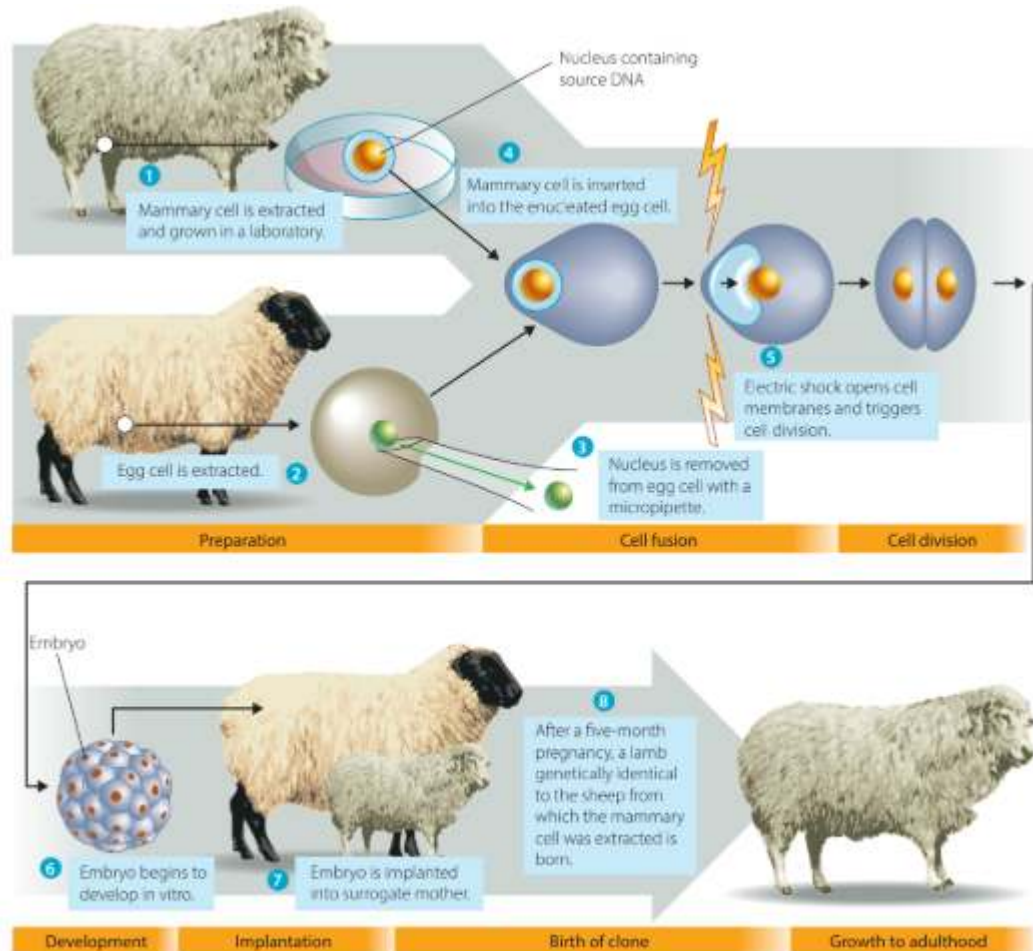
- First mammal cloned
 - Dolly the sheep
 - From mature cells of
 - Udder from
 - 6 year old sheep
- Embryo splitting
 - Occurs in nature
 - Easy to replicate in lab
- Mature cell → whole organism
 - Not natural
 - Difficult, expensive
- Main difficulty cell cloning
 - Certain genes needed for development
 - “Switched off” or “shut down”
 - In differentiated cells
 - ∴ genes need to be “switched on”
 - Or reprogrammed for development

Cloning Techniques:

Somatic Cell Nuclear Transfer:

- Scnt
- Each time a mammal is cloned
 - Process involves 3 animals
 - 1 donating nucleus
 - 1 egg doner
 - 1 surrogate mother
- Dolly the sheep process
 1. Cell from udder (mammary glands)

- of 6 year old ewe
- Sheep 1
- Cell starved of nutrients to
 - Stop them from dividing
- 2. Nucleus removed from healthy unfertilised egg
 - From other sheep
 - Sheep 2
 - Process "enucleation"
- 3. Udder somatic cell with nucleus
 - From sheep 1
 - Injected → egg cell
 - Sheep 2
 - Cells treated with electricity
 - Fuse cells together
 - To form 'fertilised' egg cell
 - Cell underwent normal growth + development
 - Dividing by mitosis
- 4. 'Embryo' implanted → uterus of 3rd sheep (surrogate mother)
 - Sheep 3
 - During when embryo cells still dividing
- Embryo born genetically identical → sheep 1
 - Udder cell sheep



Artificial Embryo Twinning:

- short time after fertilisation
- Embryo split → 2
 - Before cells become specialised
- Splitting done in lab
- Embryos implanted → surrogate
 - To develop
- Embryos that develop are genetically identical
- Technique used extensively in cattle production

Ethical Issues:

- Consider type of cloning being assessed
- Issues with whole organism + human cloning
- Animal welfare:
 - Concern
 - Treatment of animals in large scale farming
 - People may feel bad by
 - Cloning animals for food production

- Same techniques used to clone animals
 - May be used to clone humans
 - Raises
 - Moral
 - Religious
 - And legal concerns
- Religious argument that humans
 - Are acting as 'god' by
 - Cloning animals
- Unknown health risks
 - For cloned animals
- Reproductive cloning
 - Expensive
 - ∴ limiting access to technique

Population Gene and Biodiversity Issues:

- Cloning used as a form of selective breeding
 - When 'ideal' hybrid obtained
- Advantage of cloning
 - Reduces 'unknown' element in
 - Selective breeding
 - Characteristics bred can be precisely controlled
- Allows desired characteristic to be reproduced
 - In short space of time
 - Making it efficient for
 - Getting desired characteristics
 - In organisms
- Cloning reduce genetic diversity
 - In populations
 - As organisms produced by
 - Reproductive cloning
 - Derived from only 1 parent
 - (artificial form of asexual reproduction)
 - Offspring genetically identical to parent
- Only few parent animals cloned
 - If many clones from few parent organisms
 - reduce genetic diversity
 - As all organisms have identical dna

- Cloned organism
 - Becomes predominant organism
- Identical combinations of genes
 - (whole clone genomes)
 - Are conserved within population
 - $\therefore$ increase freq. Of these genotypes
 - Within population
- Due to Artificial selection
 - Natural gene combinations
 - That are not selected
 - Will gradually disappear
- Disadvantage of cloning
 - All members of species identical
 - Population less likely to
 - Survive sudden environmental changes
 - Vulnerable $\rightarrow$ foreign pathogens

Effectiveness:

- Not effective
- Low success rates
 - Improved for cattle especially
- Clones have developmental defects

Recombinant DNA Technologies:

- Insert gene from 1 species
 - Into genome of another
- Process:
 1. Gene isolated from cell
 2. Piece of circular dna (plasmid)
 - Removed from bacteria
 3. 2 pieces of dna cut
 - Using same restriction enzyme
 4. Fragments produced have
 - matching sticky ends
 - sections of single stranded dna
 - With exposed nucleotide bases
 - At end of a double stranded molecule
 - Sticky ends fit together perfectly

- Because of being cut
- By same restriction enzyme
- 5. Bacterial plasmid
 - Is cut at 2 points using
 - Same restriction enzymes
- 6. As Sticky ends of human gene + plasmid
 - Come together
 - They join up through base pairing
 - 'annealing'
- 7. Dna fragments joined by enzyme
 - Dna ligase
 - Joined fragments from
 - Circular plasmid
 - Linear molecule
- 8. Plasmid inserted back → bacterial cell
 - As bacterial cells multiplies
 - many copies of gene produced
 - When multiple copies of gene produced
 - Gene inserted
 - → egg cell of another species
 - + after fertilization
 - Becomes part of the new organisms dna

Delivering the Gene:

- Micro injection:
 - of dna → single cell nucleus
 - Done under optical microscope
 - To introduce dna → egg cells
 - When making transgenic species
- Biolistics:
 - Mechanically delivering dna on microscopic particles
 - → target tissues, cells
 - By 'firing' them from 'gene gun'
 - E.g.
 - Tiny gold particles coat dna
 - Fired at target cells
 - Under high pressure, voltage
- Electroporation:

- Increase cell membrane permeability
- By applying electrical current
- Transduction by vector:
 - Dna → cells by
 - Viral vector
 - Adenovirus
 - Liposome
 - Bacterial plasmid
 - Vector injected → bloodstream
 - Or inhaled by aerosol
 - Nasal spray
 - Oral aerosol

Transgenic Species:

- Species made by
 - Taking gene from 1 species
 - → another species
- Gene must become part of
 - Organisms germline dna
 - If to be inherited by future generations
- Gene is
 - Inserted → germline cell directly
 - Or → fertilised egg cell
- Increase reproductive capacity of animal
 - ∴ is reproductive technology

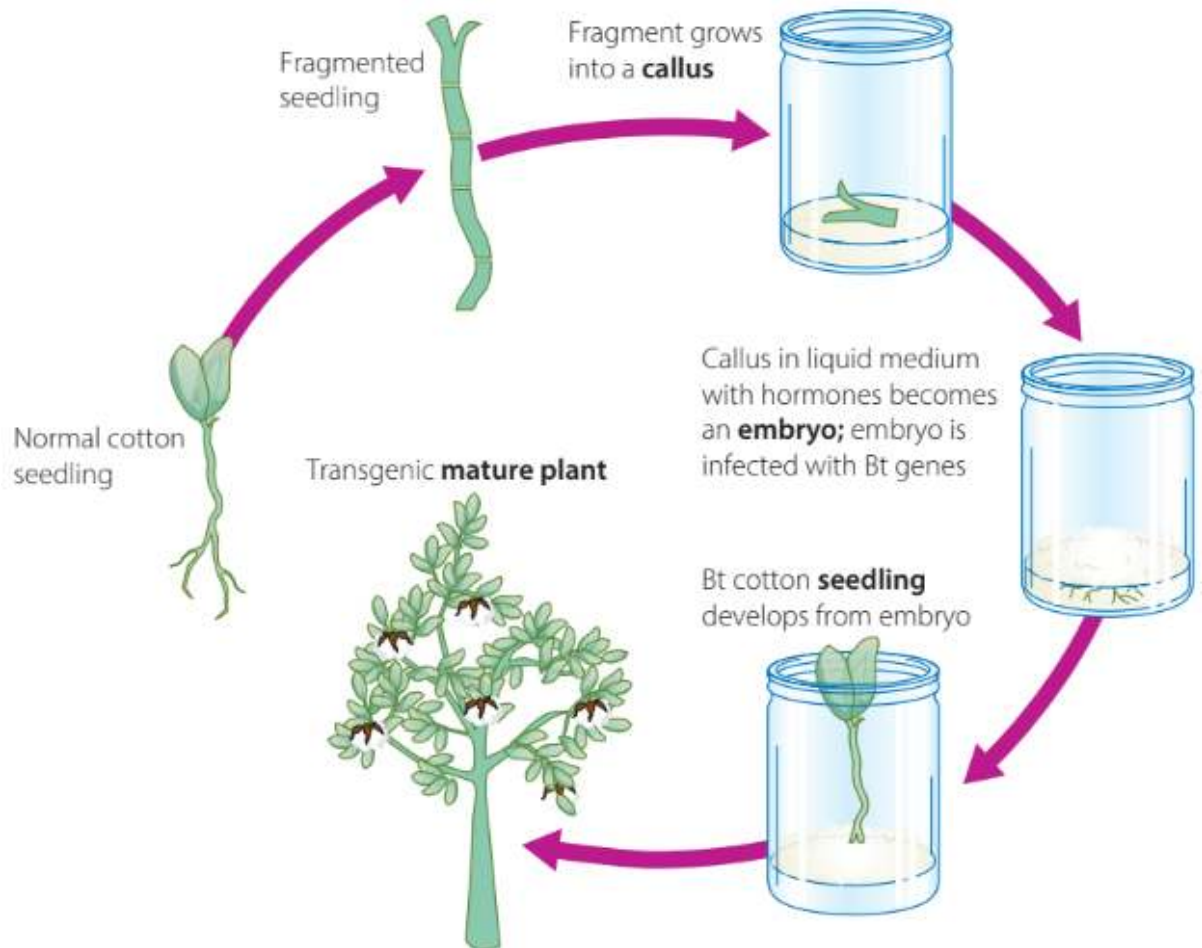
Agricultural Techniques and Applications:

Transgenic Bt cotton plant:

- Helicoverpa sea moth
 - Destroys 100 millions \$
 - Worth of cotton plants each year
- Increase pesticide
 - Moth increase resistance
- CSIRp scientists collab with US company monsanto
 - Trialled Ingard GM cotton
 - (Bt cotton)
- Bt cotton

- Genetically modified
 - (add or remove genes)
- Have gene that
 - Codes for production of protein that
 - Kills caterpillar of moth
- Insertion of bt gene → cotton plant
 - Reduced need to use pesticide
 - To kill caterpillars
 - Good for environment
 - Reduce pesticide resistance in caterpillars
- “Bt” from
 - Gene from
 - Bacillus thuringiensis
- Bt gene
 - Codes for inactive toxic protein
 - Harmless to humans
 - Animals
 - Insects
 - When in caterpillar digestive system
 - Converted → active form
 - Kills caterpillar
- Process:
 1. Cut normal cotton seedlings
 - small pieces
 - Place them → solid growth medium
 - Grow → calluses
 - When calluses
 - Transfer → liquid growth medium
 - Given hormones
 - → induce them to grow into
 - Cotton plant embryos
 2. Bt gene cut from bacillus thuringiensis
 - Using restriction enzymes
 3. Bt gene → plant embryos
 - Using vector bacteria
 - Inject genes → embryo cells
 - Dip embryos in solution

- With bacteria vector + Bt gene
- 4. When gene inserted
 - Embryos with Bt genes
 - Grown in tissue culture
 - Placed in solid medium
 - Germinate → small plants
 - Plants in pots
 - + grown in greenhouses



Golden Rice:

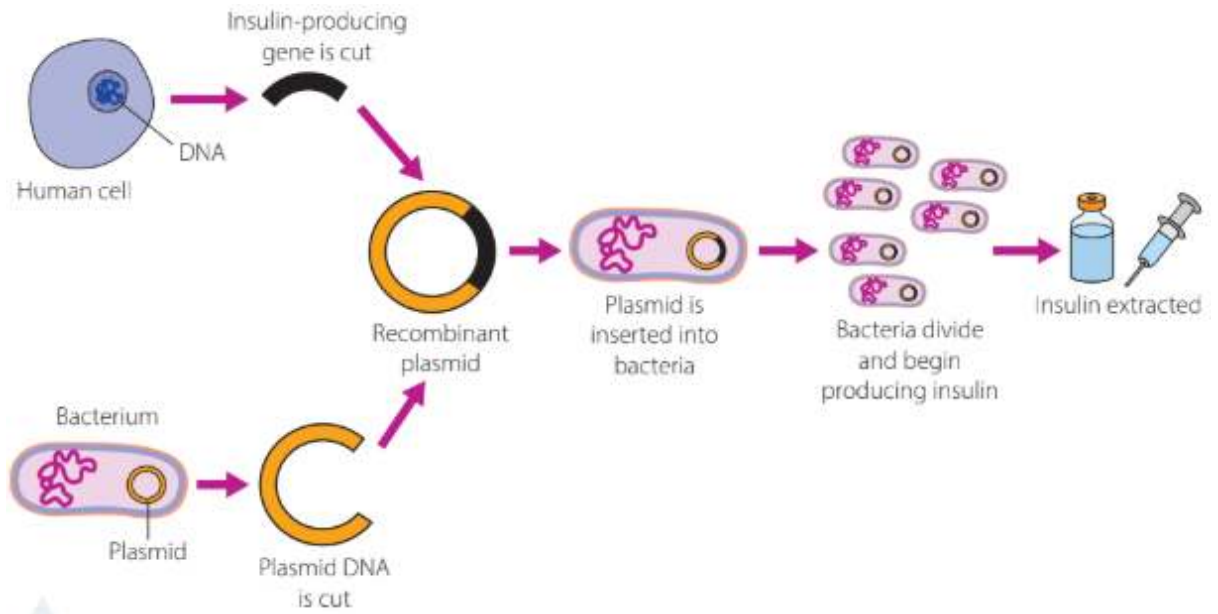
- process:
 1. Identify vitamin A gene
 - In daffodils
 - Cut using restriction enzymes
 2. Use pcr machine to
 - Make copies of gene
 3. Cut many plasmids
 - from soil agrobacterium T
 - Using same restriction enzymes

4. Paste vit A gene → plasmids
 5. Use heat shock
 - To insert plasmids → soil agrobacterium T
 6. Dip rice embryos in
 - Transgenic soil agrobacterium T
 7. Bacteria infect rice embryos
 - Insert vit a gene from plasmids
 - → embryos
 8. Rice embryos
 - Grow → rice plants that
 - Produce higher values of vitamin A
- Agriculture field use:
 - Increased nutritional value
 - Given to people in third world countries
 - To relieve hunger
 - + reduce incidences of blindness

Medical Techniques and Applications:

Insulin Production Bacteria:

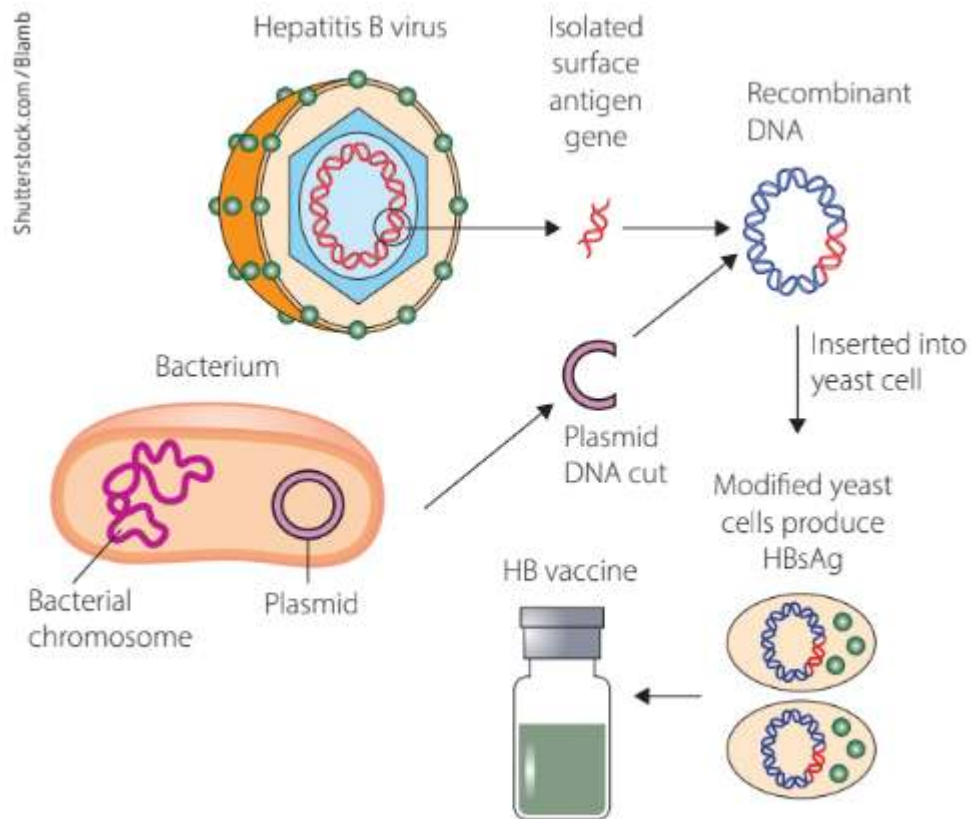
- process:
 1. Identify insulin gene from human
 2. Isolate gene by
 - Using restriction enzymes to cut
 3. Make multiple copies of gene
 - Using pcr
 4. Use same restriction enzyme to
 - Cut bacterial plasmids in e coli
 5. Use dna ligase to
 - Put genes → cut plasmid
 - Make recombinant dna plasmids
 6. Add promotor + regulator seq.
 - → gene
 - Allow it to be expressed
 7. Use heat shock to
 - Insert recombinant plasmid
 - → bacteria
 - Create transgenic bacteria



-
- **Medical field use:**
 - Bacteria multiples
 - Produce many copies of same gene
 - **Produce human insulin**
 - From harvesting it from bacteria
 - Used in medication to
 - Treat people with
 - Type 1 diabetes

Hepatitis B Vaccine Bacteria:

- process:
 1. Select **gene for**
 - **Antigen of hep b virus**
 2. Gene cut restriction enzyme
 3. Bacteria Plasmid cut same restriction enzyme
 4. Gene + plasmid
 5. **Gene + plasmid → yeast cell**
 6. Yeast cell multiplies
 7. **Antigen extracted + purified**
 - **For use as vaccine**
- Medical field uses:
 - **Treat hep B disease**
- Effects:
 - Low side effects
 - Relatively cheap to produce
 - Good for use in developing countries



Benefits of Using Genetic Technologies:

Evaluating Benefits:

1. Define all subjects
 - (genetic technology)
2. Write points for and against
 - With examples
 - To provide support
3. Make judgement on how effective tech is
 - For improving
 - Human life
 - Animal health
 - Environment
 - biodiversity

Agricultural Benefits:

- Producing crop + animal varieties
 - That are better suited to
 - Specific environments
 - E.g. high salinity, drought
- Plants made **pest resistant**

- For producers:
 - Use of transgenic organisms
 - provides opportunity to
 - Increase productivity of marginalised land
 - + reduce after harvest losses
- Advances in production of
 - Gm plants
 - With enhanced nutrient levels
- Plants gmed to
 - Have higher
 - Protein
 - Iron
 - Lipid content
 - Useful for animal feed, fuel, oil
 - + industrial applications
- Gm animals used as food sources

Medical Benefits:

- Genomics
 - Influence medical care
 - Individualise treatments
 - Used currently in
 - Oncology
 - Pharmacology
 - Treatment of rare, undiagnosed diseases
 - + diagnosed infectious diseases
- Produce valuable products
 - Through recombinant dna technology
 - E.g. humulin
 - Human insulin
 - Produced to regulate human insulin levels
 - Recombinant form is
 - Better tolerated by diabetics
 - Over previously used pig + cow insulin
- Monoclonal antibodies:
 - Cancer research
 - Artificially clone antibody producing cells that
 - Target specific antigens

- Used in cancer treatment to
 - Assist natural immune system to
 - produce specific antibodies that
 - Target cancer cells
- Grown in
 - Vitro
 - Mice stomach lining
- Recognize specific types of
 - Protein markers on cancer cells
- Produced for specific forms of cancer
 - ∴ antigen specific

Industrial Benefits:

- Gm plants may
 - Be used to produce environmentally friendly
 - Chemicals that replace
 - non renewable products
- enzymes
 - Recombinant dna techniques used
 - To make enzymes that are
 - Used in food industry
 - Dairy production
 - Brewing
 - Can be quickly cloned
 - To produce large quantities
 - Of an enzyme that is
 - Targeted at specific substrate
 - Are purer than traditional methods
 - Which have range of additional chemical substances
- developing bacteria genetically engineered
 - Plants + bacteria
 - E coli
 - That can absorb heavy metals from
 - Contaminated sites
 - Other heavily polluted areas
 - Reducing harm
 - → local ecosystems

Effect of Biotechnology in Agriculture:

Evaluating Effect:

- Define biotechnology:
 - As involving use of living systems + organisms
 - To develop or make products
 - Can have pos + neg impact on genetic variation of
 - Individual
 - Populations
 - species
 - Reduce genetic variation
 - Lessen species resilience to
 - Changes in environment + disease
 - Also reduce environmental ability to
 - Meet future human needs
- Define a biotech
- What is the biotech used for
- Pos + neg effects of biotech
- Make a judgement
 - Biotech has → reduction or increase in biodiversity
 - Should be used as it is vital to
 - provide enough food for growing population
 - + can be used responsibly to
 - Maintain + increase biodiversity
 - To continue safe use of biotech
 - Implement ethical guidelines,
 - Biosafety protocols
 - Clear legislations for
 - Genetic + animal manipulation
 - To prevent damage to biodiversity

Reproductive Technologies:

- Types:
 - Artificial selection,
 - Insemination,
 - Pollination

Advantages:

- E.g. Breeding of cattle
 - Resulting in great milk output
 - + economic gain
- Artificial insemination, pollination:
 - Introduces new genes to a population
 - May short term increase genetic variation in gene pool
- Whole organism cloning, gene cloning:
 - Reintroduce lost genetic variation
 - Into a population
 - Restore ecosystem function
 - Restore genetic resilience
 - Increase populations of species going extinct
- Bt cotton:
 - Decrease harmful environmental impact
 - normal agriculture has on environment

Disadvantages:

- Reduce genetic variation
- Due to
 - Repetitive selection of
 - specific desired genes
- E.g. Breeding of cattle
 - Threatened other native cattle
 - As because of their advantageous genes
 - They outcompete native species
- Whole organism cloning:
 - Multiple identical genetic copies
 - Becoming widespread in a population
 - → loss of natural vigour
- Risk of pesticide + insect resistant species
 - → invasive pests in natural environment
- Reduction of non target species
 - like honeybees

Genetically Modified Organisms - Bt Cotton:

Advantages:

- Introduce new alleles
 - → population
 - Immediately Increase genetic diversity
- Results in
 - Reduced use of pesticides
 - Reduced harm on surrounding plants, animals
 - Increase biodiversity

Disadvantages:

- Continuous selection of this variety
 - → loss of other varieties
 - Reducing genetic diversity
- Reduces species diversity by
 - Reduce caterpillar population
 - + other organisms that
 - Feed of caterpillar
- Outcompete other variants
 - → loss of genetic diversity

Social, Economic and Cultural Influences on Biotechnologies:

Social:

- Physical + social setting where we live
- Biotechnological techniques
 - Based on
 - specific needs of that society
 - Choices made by government
 - Wealth of individuals
 - Economic status of country
- E.g. dna fingerprinting techniques used in
 - forensic science
 - Paternity testing
 - In techniques
 - Non coding dna segments used to
 - determine differences

- + similarities
- Most common technique
 - Short tandem repeat (str) dna analysis
- Ethical issues may come from
 - Using biotechnologies

Economic Context:

- Pricing, money
- Gm foods patented
 - Mean that biotech company owns rights to
 - Particular biotech
- Large scale companies get a monopoly
- Small scale farmers
 - Cant afford to buy seeds for genetically modified crops
 - Continues unequal distribution of wealth
 - Between developed, developing nations

Cultural Context:

- Dependent on shared meanings,
 - Ideas
 - Beliefs of people
 - Who make up a society
- Values, religious, moral beliefs
 - + educational background of group of people
 - May influence their opinions + whether they
 - Are open to using biotechs